



Learning subject content through a foreign language should not ignore human cognitive architecture: A cognitive load theory approach



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ABSTRACT

Several widely implemented educational approaches aim to provide academic content in a foreign language. While Content and Language Integrated Learning (CLIL) works because it focuses both on content and on foreign language learning, approaches aiming at transmitting academic content through a foreign language should not be implemented without explicit foreign language instructional support. Based on cognitive load theory, there are theoretical reasons to hypothesize that exposure to new content in a foreign language without any foreign language instructional support may interfere with rather than facilitate learning compared to learning language and content separately. In three experiments conducted in higher education, a text was presented to 294 students in three different conditions: native language, foreign language, and foreign language with a translation into the native language. We varied the foreign language (English, German) and the domain (Law, Computer Science). Our results indicated that reading in the foreign language was never the best condition to learn either language or academic content. We concluded that considerable care should be taken when transmitting academic content in a foreign language, without explicit foreign language instructional support.

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1. Introduction

Around the world, and in Europe in particular, there are several approaches which aim to teach academic content through a foreign language in primary or secondary schools (immersion). Whereas some approaches aim to teach only content, others aim to teach both content and language simultaneously (variously known as Content and Language Integrated Learning - CLIL, Content Based Instruction - CBI). In France, according to the Ministry of Education in 2014, 10%¹ of children participated in such programs. In the context of growing internationalization, higher education has followed the trend initiated by primary and secondary education and also has provided tuition in languages other than the official language of the country, despite that language being neither the

learners' nor the teachers' native language (mainly English as a Medium of Instruction, or EMI). For students who did not follow a CLIL program in school, the shift to learning content in a foreign language in higher education is sudden (Arkin & Osam, 2015) and widespread (Bruton, 2013).

CLIL is defined as: "a dual-focused educational approach in which an additional language is used for the learning and teaching of both content and language (...)" (Coyle, Hood, & Marsh, 2010, p. 1). CLIL thus contains a strong pedagogical component aiming to support foreign language learning. Pedagogical approaches which aim at transmitting content through a foreign language in higher educational contexts tend to omit this important element, although their implementation is mostly based on research results from secondary programs using CLIL (Tatzl, 2011; Jimenez-Munoz, 2016). While considerable research has been carried out in this field (Bruton, 2011, 2013; Cenoz, Genesee, & Gorter, 2013; Coyle et al., 2010; Dallinger, Jonkmann, Holm, & Fiege, 2016; Dalton-Puffer, 2007, 2011; Dalton-Puffer, Linares, Lorenzo, & Nikula, 2014; Fortanet-Gomez, 2013; Verspoor et al., 2015; Piesche, Jonkmann, Fiege, & Keßler, 2016; Rumlich, 2014, 2016), some researchers have also suggested that many variables that are not

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¹ <http://www.education.gouv.fr/cid87947/classes-bi-langues-et-sections-europeennes-ou-de-langues-orientales-selo-en-college.html>.

easily controlled in ecologically valid contexts such as selection of participating students, time exposure to the foreign language in and outside foreign language classes, the learning material and its presentation (Dale & Tanner, 2012), students' and teachers' motivation, students' initial language proficiency, and the official languages of the country may lead to an overestimation of academic content and foreign language learning benefits (Dallinger et al., 2016). For these reasons (pedagogical component and contextual variables), it is obvious that CLIL may be difficult to transpose to every context. In particular, it should not be transposed to higher education without any foreign language instructional support, as it often occurs.

In this context, some researchers have warned against a lack of foreign language instruction in bilingual academic programs (Jimenez-MunozA, 2016) and in adjustments to the approach of already existing mother-tongue-taught courses. We may need to be concerned by what happens in situation where students, who are not selected, not particularly motivated, not particularly prepared and not particularly trained in the foreign language, are exposed to academic content in this foreign language without any foreign language instructional support. "The negative potential impact on grades and outcomes (Clegg, 2001) of this lack of adaptation is an evident concern" (Jimenez-MunozA, 2016, p.112) to researchers of the field. In this paper, we argue that empirical research using randomized, properly controlled trials (Shohamy, 2012; Perez-Canado, 2013) and theoretical argumentation, here based on cognitive load theory, are needed and can contribute to the discussion.

Our empirical work was intended to investigate some of the relevant variables that may determine the efficacy of foreign language instruction in higher educational contexts. As previously mentioned, there are reasons based on cognitive load theory (Sweller, 2015, 2016a; Sweller, Ayres, & Kalyuga, 2011) to hypothesize that without any foreign language instructional support, learning academic content through a foreign language is likely to lead to sub-optimal results. We will outline some of the available data followed by predictions flowing from cognitive load theory.

2. Learning language and content simultaneously

2.1. Definition and theoretical justification

CLIL is defined as a dual-focused educational approach, whereby a foreign language is dedicated to learning and teaching both the content and the foreign language (Mehisto, Marsh, & Frigols, 2008) in a balanced way. In Europe, CLIL, and in North America, Content Based Instruction (CBI) are widely used acronyms with no fundamental difference between the two terms (Cenoz, 2015). Such approaches thus include a pedagogical component to support both content and foreign language learning. English as a Medium of Instruction (EMI) designates certain courses or programs offered in English where English is a foreign language (Doiz, Lasagabaster, & Sierra, 2012), in particular in higher educational contexts and describes approaches in which a foreign language is used to learn content, without any or with restricted instructional support to learn the foreign language, even if the use of the foreign language in this context is likely to incidentally enhance students' competence in the second language.

Presenting content in a foreign language with no instruction concerning the foreign language whatsoever results in immersion. Immersion occurs mainly in higher education and is called EMI when the language of instruction is English. We are primarily concerned with this common version of immersion, where content

is transmitted in a foreign language without or with very limited foreign language instruction.

Many foreign language acquisition theories support positive consequences of immersion on students' foreign language skills (Dallinger & et al. 2016). The (socio)constructivist approach (Bruner, 1978), the input hypothesis (Krashen, 1985), interaction approach (Gass & Mackey, 2007), the natural approach (Krashen & Terrell, 2000) advocate that foreign language learners need sufficient amounts of input and opportunities to interact in a foreign language to learn efficiently. Moreover, according to these theories, the best learning conditions should be similar to those of native language acquisition. CLIL, EMI and immersion-classrooms conform to these criteria (Eurydice, 2006, Zydati, 2007; Surmont, Craen, Struys, & Somers, 2014).

Another theoretical argument in favour of CLIL or immersion is supported by research on bilingual children. As emphasised by Piesche et al. (2016), CLIL-students are able to "process information more deeply because they invest more mental effort in the task" (p.109). This claim is supported by the work of Heine (2010) who promoted a cognitive linguistic perspective on CLIL and showed that a focus on language has positive effects on the processing of semantic content. In sum, this stream of research suggested that cognitive control and selective attention can be intensified through processing information in a second language, and that this procedure fosters better long-term retention. Piesche et al. (2016) also emphasised that other theories indicate the opposite. For example, from a cognitive load theory perspective, (Sweller et al., 2011), students' working memory can be overloaded by simultaneously processing new content and the foreign language.

2.2. What do the data say?

Grounded on the previous theoretical considerations, several studies have been concerned with the effect of CLIL instruction on students' language skills (Jexenflicker & Dalton-Puffer, 2010; Ruiz de Zarobe, 2010) and have shown positive effects of this approach on linguistic outcomes (Admiraal, Westhoff, & de Bot, 2006; Lasagabaster, 2008; Ruiz, de Zarobe, 2008; Loranc-Paszylk, 2009; Várkuti, 2010). Dalton-Puffer (2007, 2011) suggested that students have better language performance, and show more ability, accuracy and fluency in using a foreign language than mainstream students. Ikeda (2013) indicated that a 'weak' form of CLIL works in the Japanese context if teachers are trained properly. Jiménez-Catalán and Ruiz de Zarobe (2009) also reported advantages in favour of CLIL students in the development of reading comprehension skills. In contrast, Roquet and Pérez-Vidal (2015) investigated the differential effects of two learning contexts (formal language instruction as opposed to content and language integration) on the written production skills of adolescents but the superiority of CLIL could not be confirmed. Pladevall-Ballester and Vallbona (2016) did not find any significant differences between CLIL and non-CLIL students regarding reading skills. Rumlich's longitudinal study results in Germany testing almost 1400 German students were quite disappointing and "counterintuitive" (Rumlich, 2016, p. 448) concerning CLIL benefits on learners' performance (see also Rumlich & Stebner, 2016), although motivation increased for language learning among prospective CLIL students (Rumlich, 2014), no significant effect of CLIL was found. Dallinger et al. (2016) found that a CLIL approach increased listening comprehension but no other language skills.

Concerning the learning of content, the results are even less conclusive and some studies demonstrated little benefit for content learning (Van de Craen et al., 2007) while others (Badertscher & Bieri, 2009; Dallinger et al., 2016; Jäppinen, 2005; Piesche et al.,

2016) showed no benefits suggesting that CLIL students were neither better nor worse than their non-CLIL counterparts. Marsh, Hau, and Kong (2000), studying the effect of late immersion in Hong Kong, found negative effects of instruction in English on learning, therefore contending that these findings challenged the implementation of such approaches. In sum, the available results are contradictory and the CLIL approach has generated much discussion and debate among researchers (see Bruton, 2013; the reply of; Hüttner & Smit, 2014; or; Cenoz, Genesee, & Gorter, 2013; the reply of; Dalton-Puffer et al., 2014).

These heterogeneous results led Bruton (2013) to the quite ironic conclusion that the fundamental argument for simultaneous content and foreign language learning relies on a “two-for-the-price-of-one” (p. 588) perspective, which is often very attractive but can result in unfortunate, unintended consequences. Concerning the learning of content, for example, Bruton suggested “it would be possible to have the opposite, 1/2-for-2” result (Bruton, 2013, p. 594). For him a consequence is also that: “[w]here CLIL has been adopted, it has diverted much of the attention away from the so-called mainstream school FL [foreign language] teaching” (Bruton, 2013, p. 595). In the present study, we assume that CLIL can be effective when it includes foreign language instructional support, but we are concerned with what happens in a context where mainstream students, who are not selected, not particularly motivated, and not specifically trained in the foreign language, have to learn content through a foreign language without any foreign language instruction.

2.3. Selection bias in CLIL-research

On a methodological level, the equivalence of students used to compare CLIL and non-CLIL students is important. Students and researchers (Dallinger et al., 2016) have argued that in CLIL contexts, “learning is most strongly influenced by students’ characteristics, such as prior achievement, general cognitive abilities, motivation, or demographic factors, followed by classroom factors, i.e., instructional quality or classroom composition (Opdenakker, VanDamme, DeFraine, VanLandeghem, & Onghena, 2002). Due to selection mechanisms in CLIL-programs, both student- and classroom-level factors might systematically differ between CLIL- and non-CLIL-classrooms and therefore confound the CLIL-effect” (p. 24). Aware of this issue, Gablasova (2014a) selected students from a CLIL school rather than from a non-CLIL programme as a control group “in order to control for factors such as academic performance, socio-economic status and motivation which are usually higher among the students in bilingual than in mainstream programmes” (Gablasova, 2014a, p. 154). In contrast, using non-CLIL students as a control group, as has occurred in much work (Admiraal et al., 2006; Zydati, 2007; Alonso, Grisaleña, & Campo, 2008; Lasagabaster, 2008; Ruiz; de Zarobe, 2008; Lo & Murphy, 2010), is not comparing like with like, because, as indicated by Gablasova (2014a), CLIL and non-CLIL students are different (see also Rumlich, 2016). They do not have, for example, the same time exposure, the same practice of the foreign language, the same average general level or the same motivation for learning languages.

In a similar vein, Bruton (2013) argued that some of the benefits of CLIL can be attributed to factors other than those linked to academic content presented in a foreign language. We can assume that CLIL sometimes works because students are selected (Bruton, 2011) to participate in these programs and because they are intensively trained in the foreign language. They are therefore able to deal with more difficult tasks than mainstream students, or because their

teachers are more engaged (Bruton, 2013). According to Rumlich (2016), teacher- and lesson-related variables must be taken into account to give a more accurate picture of simultaneous content and foreign language learning, as well as class composition, preparation and selection “which introduce an undesired source of additional systematic variance into data from CLIL students” (Rumlich, 2016, p. 446). Yamano (2013) divided absolute beginners in English into CLIL and non-CLIL classes in a Japanese primary school and his research suggests that CLIL can “contribute to increased student engagement and learning of content and language from the very onset of a CLIL course” (p. 28). However, in the non-CLIL class, the traditional procedure PPP (Presentation, Practice and Production) was used whereas the CLIL class provided a more meaningful learning environment with the use of coloured clay. Rumlich (2016, p. 448) advises we should use statistical approaches “that capture as many of the relevant influences as possible in order to obtain the most realistic picture of the effects of CLIL environments”. To be able to appraise the real benefit of simultaneous content and foreign language learning, Rumlich (2016) suggested the need for a “true control group”, “that is not significantly different from the CLIL group on variables determining proficiency or affective-motivational dispositions” (p. 445), otherwise, “it is impossible to separate the effects of CLIL from those caused by the implementation of CLIL” (p. 446). We strongly agree. While ecologically valid studies can be very valuable, we need at least some studies that use experiments with a strict control of variables. Without such studies, we never can be sure whether differences between conditions are actually due to the hypothesized factors. This issue provides the primary rationale for the experiments of this paper. We chose to test a common approach which is different from CLIL, but finds its origin in the success of CLIL, and consists in transmitting academic content through a second language with no or very limited second language instruction and support.

For all these reasons, we claim that any oversimplified transposition of approaches aiming at transmitting academic content in higher education without any instructional/pedagogical support on foreign language learning has to be carried out very carefully. Research on EMI (or on foreign languages other than English as a medium of instruction) provides insights into the challenges (Airey, 2004, 2012) that teachers and students have to face when a non-native language becomes the medium of academic content instruction, without language assistance. Researchers in EMI have made it clear that it is difficult to learn and to teach content through a foreign language without supporting L2 learning. Content lecturers, lacking competencies for solving language-related issues, sometimes oversimplifying content to facilitate student comprehension (Costa & Coleman, 2010; Hajer, 2000). It is also a challenge for students whose foreign language competencies do not allow them to follow the lecture (Erling & Hilgendorf, 2006). Jimenez-Munoz (2016) indicated that pedagogical adjustment such as language support and student-led tasks have an impact on students’ grades in such contexts.

2.4. Cognitive load theory

When considered from a cognitive load theory viewpoint, there are reasons to suppose that simultaneous learning of content and a foreign language is likely to be counterproductive. We will outline relevant aspects of the theory.

Cognitive load theory is based on an evolutionary view of human cognitive architecture and that evolutionary view suggests instructional procedures vastly different from the simultaneous

content and foreign language learning perspective. The theory uses Geary's distinction between biologically primary and secondary knowledge (Geary & Berch, 2016; Geary, 2012; Sweller, 2016b; Sweller et al., 2011). We have evolved to acquire primary knowledge over countless generations. Because we have specifically evolved to acquire such knowledge, our learning in primary areas tends to be effortless, automatic, unconscious, and does not require explicit tuition. For present purposes, learning to listen to and speak our native language provides an example. We have evolved to acquire a native language effortlessly, automatically, unconsciously, and without tuition simply by membership of a society.

Biologically secondary knowledge is needed for cultural reasons but we have not specifically evolved to acquire secondary skills. Unlike primary knowledge, secondary knowledge requires conscious effort and is assisted by explicit tuition (Sweller, Kirschner, & Clark, 2007). We theorise that learning a foreign language as an adult provides an example of the acquisition of biologically secondary knowledge. We may need to learn a foreign language for cultural reasons and, of course, we are able to learn a foreign language, but we can hypothesise that we have not specifically evolved to learn a foreign language and the cognitive machinery and procedures used are vastly different from the machinery and procedures used to learn a native language. The results of those differences can be seen by anyone who lives in an immigrant society. Such societies exemplify numerous cases of uneducated adults who are never able to fully learn to understand or speak the new language despite immersion in the relevant culture. Their young children on the other hand, rapidly acquire native fluency. Of course, many will learn the foreign language. We speculate that educated adults will use their skills to consciously and deliberately learn the foreign language as biologically secondary information in a manner very different to how they learned the biologically primary information of their native language.

We suspect a major reason for the emphasis on CLIL or EMI has been a tendency to ignore the above evolutionary psychology. If learning a foreign language is no different to learning a native language, then the same procedures can be used for both. We acquire a native language by immersion at the same time as we acquire content knowledge. For a native language, just as for all biologically primary skills, *two-for-the-price-of-one* really does apply. We can and do acquire a native language while simultaneously acquiring all of the other skills necessary to our lives. We have evolved to do so. But we have not evolved to acquire multiple, biologically secondary skills simultaneously. The cognitive architecture relevant to acquiring biologically secondary knowledge, discussed next, is designed to render it difficult to acquire large amounts of secondary knowledge simultaneously. That cognitive architecture can be described by five basic principles (Sweller & Sweller, 2006; Sweller, 2015; Sweller et al., 2011) that apply equally to biological evolution (Sweller & Sweller, 2006) and so constitute a natural information processing system.

2.4.1. *The information store principle*

Humans have evolved to acquire an immeasurably large amount of information that is stored in long-term memory. That information consists of both biologically primary and secondary knowledge.

2.4.2. *The borrowing and reorganizing principle*

The huge amount of information stored in long-term memory is largely acquired by imitation in the case of biologically primary information (Bandura, 1986) without explicit instruction. In contrast, biologically secondary information is acquired mainly by listening to what others say and reading what they write.

Accordingly, we have evolved to acquire a native language by unconsciously imitating other people. We acquire a foreign language by conscious effort and explicit tuition in either spoken or written form.

It should be noted that our ability to acquire both primary and secondary information from others is itself a biologically primary skill. We have evolved to acquire information from others and so do not need to be taught to acquire that information, although we do need to be explicitly taught the biologically secondary skills of reading and writing either a native or a foreign language.

2.4.3. *The randomness as genesis principle*

When we do not have information from other people, we can still obtain information using problem solving. Ultimately, all problem-solving strategies designed to generate novel information rely on random generation of problem-solving moves followed by tests to ascertain the effectiveness of the moves in attaining a problem goal, with effective moves retained and ineffective moves jettisoned – a random generation and test procedure.

Active, conscious problem solving is used in the acquisition of secondary knowledge such as learning a foreign language. Problem solving is used when the borrowing and reorganizing principle cannot be used because suitable sources of information are unavailable. For example, considerable thought may need to go into verbal expression when talking or writing with many errors that require correction. In contrast, because we have evolved to acquire primary knowledge, active, conscious problem solving is less important when acquiring a native language. As was the case for the borrowing and reorganizing principle, while the randomness as genesis principle deals with biologically secondary knowledge, the principle itself reflects a biologically primary skill that does not need to be taught because in the absence of knowledge, we have evolved to automatically engage in random generation and test.

2.4.4. *The narrow limits of change principle*

The randomness as genesis principle has structural implications. If we have three elements of information that we need to combine in some way, there are $3! = 6$ possible permutations. If we have ten elements of information there are $10! = 3,628,800$ permutations. Attempting to find a problem solution amongst 3,628,800 permutations is likely to be counterproductive. Our cognitive system has solved that problem by severely restricting the capacity and duration limits available to process novel information. As is well-known, we can process no more than 3–4 elements of novel information in working memory (Cowan, 2001; Miller, 1956) and hold that information for no more than about 20 s without rehearsal (Peterson & Peterson, 1959).

These limits apply only to novel, biologically secondary information. Accordingly, working memory is severely limited when learning a foreign language with its biologically secondary information and so the instructional effects generated by cognitive load theory should apply directly to learning a foreign language. In contrast, the working memory limits are likely to be largely irrelevant to biologically primary information that we have evolved to acquire. It follows that the working memory limits that apply to learning a foreign language are unlikely to apply to learning a native language, providing a major reason for hypothesizing that the mechanisms associated with learning a foreign language are distinct from those required to learn a native language.

2.4.5. *The environmental organizing and linking principle*

This principle provides the major justification for the preceding principles and for the entire cognitive system. The working memory limits that flow from the narrow limits of change principle

disappear when dealing with organized information stored in long-term memory. There are no known limits when working memory deals with information transferred from long-term memory. When suitable signals are received from the environment, large amounts of information can be transferred from long-term memory into working memory and used to generate action that is appropriate to that environment. The transformative effects of education derive from this principle.

Once linguistic information, either associated with a biologically primary native language or a biologically secondary foreign language has been stored in long-term memory via the information store principle, elements of that information appropriate to the context can be transferred into working memory using the environmental organizing and linking principle, organized as required and used to both understand what is being said and to generate appropriate speech. Until that information has been stored in long-term memory, neither listening nor speaking can be used effectively. For this reason, the foreign language instructional component of CLIL, which is often missing in higher education and which aims to support second language learning while learning content, is crucial.

This cognitive architecture, consisting of the distinction between biologically primary and secondary knowledge and the above principles governing the processing of secondary information, provide a base for the instructional design effects of cognitive load theory. Those effects should be directly applicable to the procedures required when learning a foreign language.

3. Implications of cognitive load theory for foreign language learning

Considering dual-focused, simultaneous learning of content and a foreign language instructional methods from a cognitive load theory perspective, we argue that such integrated approaches or learning tasks can challenge the learner's working memory in ways that have not been highlighted in previous studies. Such approaches involve acquiring two kinds of novel, biologically secondary knowledge (foreign language and subject-specific content knowledge) simultaneously. Cognitive load theory suggests that the acquisition of new, biologically secondary knowledge "tends to be conscious, relatively difficult and effortful" (Sweller, 2015, p. 2) even when presented in a biologically primary native language. Recent simultaneous content and foreign language learning research comes to the same conclusions suggesting a disadvantage of simultaneous content and foreign language learning (Piesche et al., 2016, p. 114): "Simultaneously processing content and language might have put an overload on [...] working memory capacity leading to smaller learning gains (cf. cognitive load theory, Sweller et al., 2011)". Biologically secondary knowledge cannot be acquired by simple immersion. It is acquired consciously and deliberately. If learning to speak, understand and read a foreign language as an adult are biologically secondary tasks, the difficulty of dealing with simultaneous content and foreign language learning should be consequently increased compared to dealing with both tasks independently.

However, this dual-task approach has been identified by governments and by the European Commission as an effective opportunity for pupils and students to use their new foreign language skills now, rather than learning them now for later use (see: Communication from the Commission to the Council, the European Parliament, the Economic and Social Committee and the Committee of the Regions - Promoting Language Learning and Linguistic Diversity: an Action Plan 2004–2006). Consequently, research

concerning how content is learned using a foreign language is not just an educational debate, it also constitutes an important political debate.

In the following three experiments, we would like to make clear, that in order to ensure a strict control of variables, we have deliberately foregone using long-term studies and students who have volunteered for a particular teaching procedure. We have limited the study solely to a basic approach of CLIL: learners have to read a text in a foreign language and learn about the content to be able to answer subsequent questions without access to the text. We wanted to test the hypothesis that teaching academic content through a foreign language may decrease learning outcomes.

4. The present study

Following cognitive load theory, we hypothesize that the cognitive load of a reading task in a foreign language, which aims at acquiring new knowledge in a specific field through a foreign language, can be considered as a challenge for the limited capacity of working memory (Badddeley, 2002). Researchers of the field of bilingual education often assume that written sources are likely to support knowledge and language development and that academic "reading often plays a central role in acquiring both subject knowledge and language" (Gablasova, 2014b, p.: 977). The present study compared 3 experimental conditions in which French students, who were divided into 3 groups with the same average scores on a language pretest, had to read a subject specific text (1) in French (control condition), (2) in a foreign language, German or English (without any instructional support) and (3) both in the foreign language (German or English) and in French, the mother tongue (that included limited instruction in the foreign language). In the third condition, the sentences were translated and alternated between the two languages and some critical words and phrases were underlined, highlighted in bold, or linked with corresponding translations by arrows. Students of all three experiments were attending content-related language courses in their respective universities.

The third condition in which translation from the foreign language to the native language was provided is based directly on cognitive load theory. It assumes that learning a foreign language is a biologically secondary task that depends heavily on the borrowing and reorganizing principle and so requires explicit presentation of information rather than the implicit, automatic learning that occurs when learning a biologically primary native language. The explicit presentation of information when dealing with biologically secondary information reduces working memory load (Sweller et al., 2007). This condition also corresponds to the definition of CLIL, which includes both content and foreign language instruction, even if in our case and for experimental reasons, the instruction was intentionally limited and consists only in providing a translation in the mother tongue.

The posttest consisted of:

- (1) A language test consisting of the translation of some of the identical foreign language words and phrases from the learning materials into the mother tongue. Translation was used to check whether the meaning of relevant foreign words and phrases had been learned.
- (2) A content test consisting of questions asked in the mother tongue. Questions were asked in the mother tongue to ensure that only content and not language skills were being tested at this point. An incorrect answer in the foreign

language does not indicate whether there is a knowledge gap in the foreign language or in the content.

This choice raises the question of the use of L1 and L2 in bilingual contexts, which remains a subject of debate amongst teachers and researchers of the field (see Lasagabaster, 2017). Teachers “usually fall back on their student’s other languages whenever disciplinary issues arise, to make interlanguage comparisons and to overcome comprehension problems” (Lasagabaster, 2017, p. 264). Concerning assessment of content knowledge in bilingual contexts, there is a “diversity of opinions” (Lasagabaster, 2017, p. 264; Gablasova, 2014a) about which language of assessment should be used (mother or foreign language). If the foreign language of instruction is chosen, teachers may be “sometimes unable to determine whether it is the lack of knowledge or language skills that prevents students from demonstrating their knowledge” (Gablasova, 2014a, p. 152). If the mother tongue is chosen, students taught through a foreign language may have difficulty expressing their knowledge in their native language because they have to translate, for example, specific terminology, which might not be available in the mother language, although they have understood its meaning in the foreign language. Gablasova (2014a) indicates that different aspects of content knowledge are emphasized depending on the language of assessment. When answering in the mother language, students taught in the foreign language “appeared unaware of the correct disciplinary terms and chose more general words for expressing their knowledge.” (Gablasova, 2014a, p. 161). She underlines further that an assessment in the mother language “put bilingually-educated students at a disadvantage when they are compared to L1-educated students” (Gablasova, 2014a, p. 162).

Because the level of our participants was quite heterogeneous in the foreign language (English or German), and because we wanted to test their understanding of the content, and not confront them with the additional difficulty of expressing themselves in the foreign language, we chose to let them answer the content questions in the mother tongue. We also justify this choice because two of our experiments were exactly the same in German and in English, so it could have been problematic to compare the results of the two experiments if the posttests had been carried out in German in the first experiment and in English in the second experiment.

- (3) A translation transfer test from the foreign language to the mother tongue of similar words and phrases to those found in the learning context but from a different context. While for this third test, learners had to translate new material, to do so successfully, they had to transfer the originally learned content to a new content area providing a more difficult content test.

The general design of the 3 experiments is presented in Table 1. They were specifically designed to test hypotheses derived from cognitive load theory. Different hypotheses may, of course, be derived from different theories under experimental conditions where cognitive load theory makes no predictions.

5. Hypotheses

As indicated above, we considered language and content performance during a learning task presented in 3 different ways: in the mother tongue; in the foreign language; in the foreign language with translation in the mother tongue. We expected the results to indicate that the bilingual condition providing explicit instruction in the form of a translation of the foreign language into the native language would reduce extraneous cognitive load and so be helpful in learning both content and language. Accordingly, we hypothesized that content learning through a foreign language condition would have a reduced relative performance compared to the other two groups on both content and foreign language learning.

6. Experiment 1

The hypotheses were tested in this experiment by teaching French law students aspects of the role and functioning of the European Court of Justice. The students were divided into 3 groups who read the relevant passage in French only, German only, or German with a French translation provided.

6.1. Method

6.1.1. Participants

One hundred and two French law students from the University of Bordeaux participated in the experiment. They were between 17 and 20 years old, enrolled in Year 1 to Year 4 of the course. Their initial level in German was pretested using an online language test of the Goethe-Institute <http://www.goethe.de/cgi-bin/einstufungstest/einstufungstest.pl>. The average performance ($M = 15.9$) corresponds to the B1 level (Threshold) according to the Common European Framework of Reference for Languages (p.23–24). At this level the learner is considered as an independent user, who “can understand the main points of clear standard input on familiar matters (...) produce simple connected text (...) describe experiences and events (...) and briefly give reasons and explanations for opinions and plans. (CEFR:24). (This level is outperformed by 24% of students learning German in France at the end of high school). All students had either no or very limited experience in studying Law texts in German. After these pretests, the students were divided into three experimental groups with equal, average scores on the pretests. Other than the requirement of equal average scores on the pre-tests, assignment to groups was random. Consequently, the groups were comparable, with skilled

Table 1
General experimental design for Experiments 1, 2 and 3.

	Experiment 1 ($N = 102$)	Experiment 2 ($N = 84$)	Experiment 3 ($N = 108$)
Pretests	German Language $M = 15.9$; min = 7.5; max = 30	English Language $M = 12.7$; min = 3.5; max = 18.5	English Language $M = 10.9$; min = 2; max = 20
Domain	European Court of Justice	European Court of Justice	Assembly language
Control condition	Text in French	Text in French	Text in French
Content exposure through a foreign language condition	Text in German	Text in English	Text in English
Content exposure with limited instructional language support condition	Text in German with translation in French	Text in English with translation in French	Text in English with translation in French
Posttests	Language + Content + Transfer	Language + Content + Transfer	Language + Content + Transfer

and less skilled foreign language learners in each group.

6.1.2. Instruments

The text from which students had to learn was about the European Court of Justice, a law text available in French, German and English. This text described the institution, the part it plays in Europe, its mission, seat, number of judges and the way they are chosen, the petitioners, the main forms of appeal and the matters for which the court's judgment is solicited. The main difficulties for learners lie in remembering the different courts from which the European court must be distinguished, the different parties who can petition the court and the different forms of appeal. Generally speaking, learning the new specific law vocabulary in German and retrieving it for the language posttests (especially the transfer posttest) is a demanding task.

Learners were asked to read the text and try to learn as much of the language and content as possible. The participants had to read the text (1) in French (control condition), (2) in German and (3) both in German and French. In the third condition, the sentences were alternated between the two languages with each German sentence immediately followed by its French translation with critical words and phrases underlined, highlighted in bold, or linked with corresponding translations by arrows (see Appendix A for an English version with its French translation). The posttest required (1) a translation of some of the identical German words and phrases from the learning materials into French; (2) content questions asked in French; and (3) a translation transfer test from German to French of similar words and phrases to those found in the learning context but from a different context (see Appendix B for an English version). Before reading, we informed participants that they would have to answer questions about the foreign language and about the content of the text. Paper and pen were used for all materials.

6.1.3. Procedure

The pretest, learning phase and posttest took place during the German course. The German teacher was the experimenter. The pretest duration was about 1 h, one week before the experiment itself. The participants were not informed about their performance. The learning phase was 15 min long. Learners were given 15 min to read and understand the text. Next, the participants immediately were asked to complete the three posttests. The total duration of the posttests was 15 min. The three posttests were scored with one point per correct answer.

6.1.4. Results and discussion

ANOVAs on the language, content and transfer test scores showed an effect of the experimental condition on both the language, $F(2, 99) = 7.90$, $p < 0.001$, $\eta_p^2 = 0.138$, and content, $F(2, 99) = 18.59$, $p < 0.001$, $\eta_p^2 = 0.273$, posttests but not on the transfer posttest, $F(2, 99) = 1.39$, $p = 0.25$.

Following the two significant F tests, post-hoc comparisons were carried out to test for individual differences between means (Table 2). For the language posttest, performance was better for German with translation compared to French only, $LSD = 4.83$,

$p < 0.001$, or German only, $LSD = 7.29$, $p < 0.001$. The difference between the French only and German only conditions in learning the language was not significant, $LSD = 2.460$, $p = 0.095$.

For the content posttest, performance was better for French only compared to German only, $LSD = 5.953$, $p < 0.001$, and to German with translation, $LSD = 3.270$, $p < 0.001$. In addition, German with translation was better than German only, $LSD = 2.683$, $p = .003$.

These results are largely in accord with our hypotheses. For language learning, an explicit translation of the text provided the best test scores with the worst scores obtained by the immersion, foreign language only group. In contrast, for content learning, the best scores were provided by the native language only group but the worst scores again were obtained by the immersion group. Immersion decreased both language and content scores.

7. Experiment 2

This experiment replicated the first experiment with the same materials and procedures. The only change was that English was the foreign language instead of German.

7.1. Method

Eighty four third year French law students were tested from the same University as the students of Experiment 1. These students were learning English as a foreign language, reading the same text as in Experiment 1 about the European Court of Justice either in French, English or in English with a translation in French. Learners were approximately 20 years old. Their initial level in English was pretested according to two online languages tests (<http://www.cambridgeenglish.org/fr/test-your-english/adult-learners> and http://www.examenglish.com/leveltest/grammar_level_test.htm): The average performance ($M = 12.7$) corresponds to the B1 level (Threshold) according to the Common European Framework of Reference for Languages (this level is outperformed by 18% of students learning English in France at the end of high school). We took the average score of these two pretests as an indicator of their initial level. Apart from the switch from German to English, all protocols and procedures were exactly the same as in Experiment 1.

7.2. Results and discussion

ANOVAs on the 3 posttest scores showed no effect of the experimental conditions on language, $F(2, 81) = 1.295$, $p < 0.28$, a significant effect on content, $F(2, 81) = 13.20$, $p < 0.001$, $\eta_p^2 = 0.246$, and no effect on transfer, $F(2, 81) < 1$. Following the significant effect on content, post-hoc comparisons indicated a better performance for French only compared to English only, $LSD = 4.775$, $p < 0.001$, and to English with translation, $LSD = 2.602$, $p < 0.006$. English with translation was also superior to English only, $LSD = 2.172$, $p < 0.023$ (see Table 3).

There was no difference in the performance of the language posttest between the 3 experimental conditions. For content, the French only condition was the best. All the mean differences between the 3 conditions were significant. There was no difference in

Table 2

Mean (M) and Standard Deviation (SD) Correct Test (language, content, transfer) Score Results, depending on experimental conditions, for Experiment 1.

Posttest	Language (max = 31)		Content (max = 21)		Transfer (max = 10)	
	M	SD	M	SD	M	SD
Text in French ($N = 33$)	14.80	8.01	17.18	4.55	4.90	3.44
Text in German ($N = 35$)	12.34	7.57	11.23	4.01	5.84	3.16
Text in German with translation ($N = 34$)	19.63	7.63	13.91	4.45	5.94	2.92

Table 3Mean (*M*) and Standard Deviation (*SD*) Correct Test (language, content, transfer) Score Results, depending on experimental conditions, for Experiment 2.

Posttest	Language (max = 31)		Content (max = 21)		Transfer (max = 10)	
	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>
Text in French (<i>N</i> = 26)	25.96	2.73	18.15	3.22	6.71	1.85
Text in English (<i>N</i> = 29)	25.24	4.32	13.38	3.52	6.43	2.03
Text in English with translation (<i>N</i> = 29)	26.86	4.16	15.55	3.55	6.31	2.34

the performance on the transfer posttest between the 3 experimental conditions.

In this experiment and unlike in the first experiment we did not find any effect of the experimental conditions on the language posttest. A likely reason for the failure of a language effect in this experiment is that learners may have been more able in English than German and so did not require the same assistance and guidance when learning. Support for this suggestion comes from the much higher test scores obtained on the language test in Experiment 2 compared to Experiment 1. Indeed, there may have been ceiling effects on the language test.

There may be two reasons for this increased ability. First, there may have been a greater previous exposure to English than German because of the heavy cultural impact of English. Second, English and French are very close languages and for a French learner, the English lexicon is quite transparent with many cognates. The learners may have readily found a French translation for the English text. Consequently, in this experiment this part of our hypothesis was not supported.

To learn the content, our results show that as in the first experiment, the French only condition was the best, supporting that part of our hypothesis. Again, the immersion condition (English only) resulted in the worst scores.

8. Experiment 3

In this experiment, we replicated Experiment 2 but changed the content from Law to Computer Programming. In all other respects, the same procedures were followed as previously.

8.1. Method

The participants were 108 first-year students in computer science learning English as a foreign language for specific purposes at the University of Toulouse who read a text about a programming language, an assembly language, which was about to be taught to them the following week by computer science teachers (see [Appendix C](#) for the English version with its translation into French). This text provided a definition of assembly language, stating its specificities compared to other programming languages and gave an example of programming in this language. Common procedures of assembly language were described and the importance of the language was stressed. The main difficulty for the learners lay in understanding the characteristics of this programming language, how it operates, and in remembering the different procedures.

The posttest (see [Appendix D](#)) required (1) a translation of some of the identical English words and phrases from the learning materials into French; (2) content questions asked in French; and (3) a translation transfer test from English to French of similar words and phrases to those found in the learning context but from a different context.

The participants were approximately 18 years old. Their initial level at English was pretested using the ELAO test (<http://langues.ups-tlse.fr/test-de-niveau-elao-235021.kjsp?RH=UFRLANGUES&>) and the reading test from the “general training” IELTS ([https://](https://www.ielts.org/about-the-test/test-format-in-detail)

www.ielts.org/about-the-test/test-format-in-detail).² The average performance (*M* = 10.9) corresponds to the B1 level (Threshold) according to the Common European Framework of Reference for Languages (this level is outperformed by 18% of students learning English in France at the end of high school).

8.2. Results and discussion

ANOVAs on posttest language, content, and transfer scores showed a significant effect of the experimental condition on the language posttest, $F(2, 105) = 13.58$, $p < 0.001$, $\eta_p^2 = 0.205$, and on the transfer posttest, $F(2, 105) = 10.401$, $p < 0.001$, $\eta_p^2 = 0.165$, but not on the content posttest, $F(2, 105) = 1.014$, $p < 0.366$. Post-hoc comparisons following the two significant results indicated that for the language posttest, performance was better for the English with translation group compared to the French only group, $LSD = 2.556$, $p < 0.008$, or English only group, $LSD = 4.889$, $p < 0.001$. French only was also significant better than English only, $LSD = 2.33$, $p < 0.014$. For the transfer posttest, performance was better for French only than English only, $LSD = 1.778$, $p < 0.004$, and for English with translation over English alone, $LSD = 2.694$, $p < 0.001$. There was no significant difference between French only and English with translation, $LSD = 0.917$, $p < 0.13$ (see [Table 4](#)).

The language results of Experiment 3 are in accord with the results of Experiment 1. When learning a foreign language, the foreign language with translation to the native language was better than simple immersion in the foreign language or presentation solely in the native language. In addition, the native language resulted in more foreign language learning than presentation in the foreign language. Immersion in the foreign language resulted in the worst foreign language learning test scores.

When learning the content, there were no significant differences between conditions. The learners' knowledge of programming languages that they can transfer to a new programming language (the assembly language), independently of the condition may have determined the content posttest scores. In addition, the content test may have been too difficult for these learners.

Unlike the previous two experiments, significant differences were found on the transfer tests. Interestingly, as discussed in more detail in the General Discussion, the pattern of significance resembled to some extent the content pattern of significance in Experiments 1 and 2 rather than the language learning pattern, suggesting that this result may be due primarily to content rather than language. After seeing the content in their native French and understanding that content, it may be easier to translate the material from English into French in the post-test than after seeing the same content in English and so having difficulty understanding it (see [Table 4](#)).

² This test cannot be presented here because of copyright. It can be found on pages 143–152 in V. Jakeman & C. McDowell. 2001. *IELTS Practice Test Plus*. Harlow: Pearson Education, Longman. Examples of the General Training Reading task can be found at <https://www.ielts.org/about-the-test/sample-test-questions>.

Table 4

Mean (M) and Standard Deviation (SD) Correct Tests (language, content, transfer) Score Results, depending on experimental conditions, for Experiment 3.

Posttest	Language (max = 25)		Content (max = 15)		Transfer (max = 15)	
	M	SD	M	SD	M	SD
Text in French (N = 36)	13.42	4.40	7.78	2.95	9.06	2.89
Text in English (N = 36)	11.08	3.37	7.25	3.37	7.28	1.89
Text in English with translation (N = 36)	15.97	6.11	6.78	3.58	9.97	2.75

9. General discussion

The results of the three experiments are coherent and consistent. To facilitate comparisons, Table 5 provides a summary of the patterns of significant differences in the three experiments.

With respect to language learning, Experiment 1 teaching law in German and Experiment 3 teaching computer programming in English both indicated that teaching the native language by providing an immediate translation from the foreign language was superior to both presenting the information in either the foreign language or the native language alone. The same relations between means were found for Experiment 2, teaching law in English, but the differences were not statistically significant. These data provide no evidence that a condition, where content is taught in a foreign language provides a language learning advantage, be it incidental or “through osmosis” (Taillefer, 2013, p. 37). Language learning is facilitated by explicit teaching of the language (even if, in the present study, the instructional support is limited). This finding is in line with the claim of researchers who have pointed out the importance of instructional guidance when learning a foreign language (e.g. Jimenez-MunozA, 2016).

Not only does the presentation of content in a foreign language not provide any advantage in learning that language, it had negative consequences with respect to learning the content itself.

Table 5

Results summary.

Experiment 1

Language scores

Translation > Native Language; Translation > Foreign Language; Native Language = Foreign Language

Content scores

Native Language > Foreign Language; Native Language > Translation; Translation > Foreign Language

Transfer

No significant differences

Experiment 2

Language scores

No significant differences

Content scores

Native Language > Foreign Language; Native Language > Translation; Translation > Foreign Language

Transfer

No significant differences

Experiment 3

Language scores

Translation > Native Language; Translation > Foreign Language; Native Language > Foreign Language

Content

No significant differences

Transfer

Native Language > Foreign Language; Native Language = Translation; Translation > Foreign Language

Note: A ‘>’ sign indicates a significant difference while an ‘=’ sign indicates no significant difference.

Experiments 1 and 2 provided identical patterns of significance. Content was better learned in the native language than either the foreign language or the foreign language with translations. The foreign language with translations was better than the foreign language alone. In Experiment 3, while content scores were highest when learned in the native language, none of the means differed significantly. The content test may have been too difficult to allow significant differences to show up for this group of learners. In total and unsurprisingly, learning in a foreign language without any language instructional support provides no advantage to content learning.

The transfer scores indicated no significant differences except in Experiment 3. In that experiment, the pattern of significant differences more closely resembled the pattern for content scores than language scores suggesting that the transfer test provided a proxy for learning content. Learning in either a native language or learning with a translation into the native language was superior to learning in a foreign language. As was the case for content learning, the greater the emphasis on learning in a native language, the more content was learned, with learners able to transfer the content of what they learned to a new context.

These results are in accord with cognitive load theory. Learning a foreign language requires explicit instruction in that language with learning depressed in the absence of that instruction. And, of course, any emphasis on the foreign language is likely to interfere with content learning. Requiring learners to acquire knowledge simultaneously in two biologically secondary areas, without explicit tuition, is highly likely to overload working memory resulting in reduced learning in both areas. This is in line with findings of other researchers in the field of bilingual education: “To assist the students in learning academic subjects through an additional language, it is necessary to identify areas that may be difficult for them and to offer well-designed pedagogical support to overcome the increased learning and cognitive load” (Gablusova (2014b: 978). Treating foreign language learning as an activity analogous to native language learning may not only interfere with language learning but will simultaneously depress content learning.

The strength of the current results is that they were demonstrated in rigorously controlled experiments, reducing the likelihood of interference from extraneous variables. The weakness of this approach is that while the experiments were run in authentic environments, they were short-term. Consequently, the current results do not necessarily challenge the results of longitudinal experiments carried out across several years and with several tests, but question an oversimplified transposition of versions of CLIL that do not include explicit language instruction. Future work is required to establish whether similar results are obtainable in longer-term studies that reduce or eliminate variables that compromise the validity of any conclusions. Rumlich (2016) has initiated such work. In his long-term study in Germany he controlled variables such as selection, preparation and class composition and obtained results which indicated no significant effect of CLIL. His results are in line with the present study.

It needs to be emphasized that while standard CLIL incorporates

both content and language instruction, it sometimes implicitly is used as an umbrella term that includes immersion in a foreign language with no language learning instructional support (Taillefer, 2013). Our experiments indicate considerable deficiencies of an approach that omits explicit language support.

It is likely that with increased language or content expertise, at some point additional language skill can be accumulated by immersion without damaging content learning. The expertise reversal effect is a major cognitive load effect (Kalyuga, Ayres, Chandler, & Sweller, 2003; Sweller et al., 2011) that suggests the possibility of such a result. The effect occurs when one instructional procedure is superior to another procedure for novices but with increased expertise, the advantages first decrease and then may reverse. In the current context, detailed, intensive language instruction may be advantageous for novices. With increasing expertise in the foreign language, that advantage may first decrease and eventually, explicit instruction may become redundant, resulting in negative rather than positive effects. For learners with considerable expertise in the foreign language (above a linguistic threshold that is probably far above that of our study participants, Zydaitis, 2012), immersion may well be beneficial. This hypothesis requires testing.

9.1. Conclusions

In conclusion, based on cognitive load theory and the current results, a policy of immersion in a foreign language without explicit language instruction when acquiring content may be misguided when applied to novice learners. Advocacy for adult foreign language learning by immersion may be based on a simplistic, *two-for-the-price-of-one*, view. That view, in turn, fails to distinguish between biologically primary and secondary knowledge. For biologically primary knowledge that we have evolved to acquire, a *two-for-the-price-of-one* view is probably correct. We can acquire content and improve our knowledge of our native language simultaneously without overloading working memory because we have evolved to do so. While we can acquire a foreign language as adults, we have not specifically evolved to do so. As a consequence, the simultaneous acquisition of the foreign language and content, both of which are biologically secondary, is likely to overload working memory, especially in the absence of foreign language instruction. In contrast, cognitive load theory can be used to support immersion for advanced learners of a foreign language but that hypothesis requires testing using properly controlled studies.

The current work should not be seen as contradicting previous work using long-term, ecologically valid studies. Such studies are important but equally, studies that use a strict control of variables are also important. Our view is that there has been far too little emphasis on studies that strictly control variables and the omission of such studies can run the risk of unbalanced conclusions and inappropriate recommendations. An appropriate balance between ecologically valid and experimentally valid studies is required. Our results clearly suggest that at least under some circumstances, immersion in a second language may be inappropriate.

Cognitive load theory can provide a framework for determining appropriate instructional techniques. It suggests that evolutionary psychology and human cognitive architecture are unavoidable concomitants of instructional design. Based on the current data, ignoring these theoretical considerations may be unwise. Cognitive load theory can be used to frame research in the domain of CLIL by more accurately defining the conditions where it is possible to learn domain specific knowledge and a foreign language simultaneously. An appropriate theoretical context along with a mix of data from ecologically valid and experimentally valid studies may provide the best route to untangling the complexities of our discipline.

Appendix A. Supplementary data

Supplementary data related to this article can be found at <http://dx.doi.org/10.1016/j.learninstruc.2017.04.007>.

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